

HRIDIK HINGORANI

Curriculum Vitae

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Education

Expected May 2027 **BS Computer Engineering, Minor in Mathematics**, *University of Illinois at Urbana-Champaign*

Coursework ECE 470 Introduction to Robotics, ECE 489 Robot Dynamics and Control, ECE 486 Control Systems, ECE 391 Operating Systems, ECE 385 FPGA Design, ECE 210 Analog Signal Processing, CS 225 Data Structures (C++), MATH 257 Linear Algebra

Experience

- May 2026 – present **AI Intern**, *Software Associates*, North Potomac, MD
- Built an AI-powered algorithmic trading bot in Python. Multi-indicator signal engine (RSI, MACD, Bollinger Bands, SMA/EMA, ATR) generates weighted BUY/SELL/HOLD signals with confidence scoring.
 - Integrated Interactive Brokers Client Portal REST API for live market data, historical bar fetching, and automated order execution with session keepalive.
 - Implemented ATR-based dynamic position sizing and automated risk controls (stop-loss, take-profit, trailing stop), capping per-trade portfolio exposure at 1%.
 - Built a real-time Flask + JavaScript dashboard with auto-refreshing signal cards, live P&L tracking, open positions table, and full trade history log. Modular 6-file architecture across signal generation, risk management, and broker connectivity.
- May 2025 – Aug 2025 **AI Software Development Intern**, *HealthArk Insights*, Mumbai, India
- Built Python-based document-processing features including summarization, web scraping, and dynamic formatting tools tailored to client specifications.
 - Integrated client feedback into production workflows by gathering requirements, validating features, and refining document-management functionality. Features shipped to production.

Projects

- Jan 2024 – Aug 2025 **StethoSpy: Electronic Stethoscope**, *Project Lead, I-MADE UIUC*
- Prototype electronic stethoscope for heart-sound acquisition and murmur screening, aimed at rural and low-infrastructure settings. Custom PCB captures heart sounds; an ADC/DAC pipeline processes the analog signal to flag cardiac irregularities for follow-up. Not clinically validated. Owned hardware design and signal processing end to end.
- 8-Channel EEG Acquisition Board**, *Independent research*
- 8-channel, 2-layer EEG signal acquisition board in KiCad. Op-amp analog front end with low-pass filtering for microvolt-level biopotential acquisition; schematic capture through PCB layout designed for high signal integrity across all eight channels.
- UR3 Image-to-Motion Drawing Arm**, *Python, ROS, OpenCV*
- Closed-form analytic inverse kinematics for a 6-DOF arm, written from scratch, with product-of-exponentials forward kinematics evaluated through matrix exponentials. Drives a UR3 to draw an image: Canny edge detection and contour simplification mapped onto a 20 cm square drawing area, published as joint targets over ROS and held to 0.0005 rad per joint. A Pikachu test image came out as 142 strokes, drawn in 7 minutes.

CRS Arm Task-Space Impedance Controller, C, TMS320F28335, Control Theory

Task-space PD impedance control on a three-link arm, mapped to joints through the Jacobian transpose and closed at 1 kHz on a timer interrupt. Reads joint encoder angles and outputs joint torque commands directly to the amplifiers; it does not send position setpoints to a servo controller. Per-waypoint stiffness modes give selective compliance, with per-joint Coulomb and viscous friction compensation and torque saturation at 5 N-m per joint. Contact tasks use a feedforward downward force commanded through the motor torque constant, with no force sensor and no closed-loop force feedback, validated afterward against a physical scale. Written in C on a TI TMS320F28335. Peg insertion, zig-zag tracking, and pushing an egg without breaking it.

We Are Kernel: RISC-V OS, C, RISC-V

A RISC-V kernel from scratch, roughly 6,000 lines of C: Sv39 paging, VirtIO, a FAT filesystem, fork/exec/exit, pipes, a shell with redirection, and preemptive multitasking.

Animatronic Grogg, Python, C++, KiCad, ROS

ECE 445 senior design capstone, currently in planning. A 22-DOF bipedal animatronic. Planned scope covers the software, the LLM voice pipeline (local LLM with Claude API, Whisper speech-to-text, Coqui text-to-speech), and the mechanical design, on a custom PCB for power.

Jupiter AI Voice Assistant, Python, MLX, Qwen3, Whisper

Fully local voice assistant for Apple Silicon, no cloud and no API keys. Whisper STT, a Qwen3 router and main model on MLX with speculative decoding and a persistent prompt cache, and Kokoro streaming TTS. Continuous VAD gives barge-in that halts generation mid-sentence. 0.3 to 0.6 s first token, warm.

LQR Inverted Pendulum, MATLAB, Embedded

Inverted pendulum held upright by full-state LQR with a 3-second settling time, simulated in MATLAB and then made to stand up in hardware.

Teaching Experience

- Aug 2025 – present **Course Assistant, ECE 385 (FPGA Design), University of Illinois at Urbana-Champaign**
Supported 100+ students in SystemVerilog-based FPGA labs covering finite state machines, datapaths, memory systems, VGA output, simulation, synthesis, testbench verification, and hardware debugging. Debugged student designs during office hours and lab sections, root-causing failures in timing, reset logic, module interfaces, and FPGA deployment.
- Jan 2025 – Dec 2025 **Course Assistant, MATH 257 (Linear Algebra), University of Illinois at Urbana-Champaign**
Taught Python-based linear algebra lab sections to approximately 200 students, supporting hands-on computational exercises.

Research

- Ongoing **Project Lead, Brainwaves Team, NeuroTechX UIUC**
Leads the Brainwaves project team, overseeing the club's podcast and social media output. EEG signal-processing and neural-network research through the club.
- Ongoing **Independent EEG Research, Self-directed**
Independently designed and built the 8-channel EEG acquisition board listed under Projects; separate from the club work.

Leadership

- Jan 2025 – present **Event Lead, then Senior Coordinator, National Team, Hindu YUVA**
Manages the UIUC chapter and the event team. Organizes large-scale Hindu cultural and community events at both chapter and national scope.
- Jan 2024 – Aug 2025 **Project Lead, StethoSpy, I-MADE UIUC**
Led the StethoSpy project (see Projects), owning hardware design and signal processing end to end.

Skills

Languages Python, C, C++, MATLAB, SystemVerilog, JavaScript

- Hardware KiCad (schematic capture, PCB layout), FPGA design, analog front ends, ADC/DAC pipelines, IMU integration, SolidWorks
- Robotics & software ROS, OpenCV, control theory (LQR, impedance/force control), Flask, REST APIs (Interactive Brokers), Whisper, Coqui TTS, Claude API, LLM integration

Off the Clock

- Writing Poetry and literary fiction. The poems lean on Mumbai-specific references, Hinglish code-switching, and language kept precise. Posted, fairly regularly, on Instagram.